RAG-Legal Assist: An NLP-Driven Legal Document Processing System with Adaptive Text-Splitting for High-Precision Retrieval Legal Judgments

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Abstract

Legal documents are, by their very nature, lengthy, structurally complex and semantically rich documents, however, making the process of manual analysis and retrieval of relevant information time consuming and prone to error. Conventional keyword-based retrieval systems do not always capture the hierarchy and context of dependency of legal domains, and standalone large language models (LLMs) lack the ability to handle long contexts and can give unsupported or hallucinated outputs when working with long documents. To try to address these limitations, this paper proposes RAG-LegalAssist, which is a Retrieval-Augmented Generation (RAG) framework for question answering on a legal document that combines a novel Adaptive Hybrid Legal Chunker (AHLC). The proposed chunking strategy dynamically combines structure-aware rule-based segmentation, semantic similarity based splitting together with length constrained chunking techniques to preserve the legal boundaries while making the chunks compatible to context constraints of LLM. The system makes use of the European Court of Human Rights (ECtHR) judgment dataset that consists of some 11,000 instances that were initially available in a JSON format and later converted to a cleaned CSV corpus for easy processing. Document chunks are embedded with BGE embeddings and indexed using FAISS, whereas the generation of the answers is done using the Qwen2.5-1.5B-Instruct model. Extensive experiments were conducted on five evaluation phases like large-scale query evaluation, query type generalization, prompt engineering comparison, paraphrase robustness analysis and chunker benchmarking. Using the proposed AHLC approach, the system achieved Precision@5 = 0.40, Recall@5 = 1.00, MRR = 1.00 and nDCG@5 = 1.00 which revealed reliable retrieval performance. Generation level evaluation showed further strong contextual alignment with Groundedness = 0.8749, Faithfulness = 0.9664 and a Hallucination Rate = 0 across 50 legal queries. A prompt engineering study found a citation enclosed prompt to be the most effective prompt engineering configuration: Groundedness = 0.8799, Faithfulness = 1.00, no hallucination. Additionally, comparative analysis among several chunking strategies proved that the proposed AHLC method resulted in the highest faithfulness score (0.9333) and at the same time was able to maintain legal structural integrity in the retrieved context. The experimental results show that the proposed RAG-LegalAssist framework is a reliable and structured solution for the legal document understanding and question answering with a strong contextual grounding and minimal hallucination.

Keywords: Retrieval-Augmented Generation, Legal Natural Language Processing, Adaptive Hybrid Legal Chunking, Legal Question Answering, Vector Embeddings, FAISS Indexing, Prompt Engineering, Grounded Generation, Hallucination Reduction.

1. Introduction

The legal space generates a large amount of unstructured text data such as court rulings, statutes, contracts, legal opinions, etc. are important pieces of information for law professionals, researchers and policymakers in understanding the legal precedent to make informed decisions [5], [9], [11], [12]. With the digitization of legal records, as well as the legal phenomena of the online environment, it has become that more and more legal information is available and the increasingly felt needs for automated techniques are becoming more realistic that could efficiently process, analyze and extract relevant information from complex legal documents. Natural Language Processing (NLP) has therefore emerged as an important technology in the legal domain, enabling automated tasks such as information retrieval, document classification, summarization, and legal question answering on large-scale semantic and dense text corpus [5], [11].

First legal document processing systems were mainly based on keyword-based search methods, rule-based extraction techniques and traditional information retrieval approaches and were limited to semantic understanding of legal language [9], [11]. The development of deep learning and transformer-based models particularly made it much better for machines to determine legal text. Domain-adapted models like Legal-BERT and other transformer-based language models facilitated the enhancement of performance in tasks like classification of legal documents, summarization and question answering [5], [12]. More recently, however, Retrieval-Augmented Generation (RAG) frameworks have been introduced that can combine vector-based document retrieval with large language models (LLMs) to enable systems that can produce context-aware and grounded responses by retrieving relevant textual segments from large document collections [1], [2], [7], [10]. RAG-based legal systems have shown improvements in answer accuracy, interpretability and reduction of hallucinations, making them suitable for applications such as legal advisory systems, document analysis, legal chatbots and case law exploration tools [6], [7], [17].

Despite all these advancements, there are still significant challenges involved in the application of NLP and RAG-related systems to lengthy and structurally-complex legal documents. Legal judgements can provide examples of hierarchical structures such as articles, sections, clauses and arguments, which can be of several pages, and full of legal reasoning. Policy documents drafted in languages with highly technical legal terms have long posed a challenge for promoting the use of RAG pipelines in policy preparation, because conventional chunking approaches for RAG pipelines, such as segmentation of fixed length or naive recursive splitting of text, frequently break the semantic cohesion, as well as retaining the poll Hierarchical structure of Law documents.]. As a result, the retrieval systems may retrieve irrelevancies or incomplete chunks of the context, with a detriment to the quality of the answer. Benchmark studies have shown that the precision of retrieval is very sensitive to chunk granularity, and well segmented chunks can greatly improve the rate of hallucination and the reliability of the answers [4], [14]. Furthermore, large language models are bound by constraint contexts and can actually produce hallucinations when the information gathered is noisy or insufficient [2], [16]. Although recent research brings in improvements such as dynamic retrieval, relevance filtering, and agent-based feedback mechanisms, document segmentation is still mainly sought as a static preprocessing step to the majority of RAG systems [3], [8], [15], [17]. As a result, existing legal RAG frameworks tend to have difficulties in delivering a good and contextually correct responses when processing long judgments with complex legal structures.

In order to overcome these limitations, this work presents RAG-LegalAssist which is an NLP-based legal document processing system that incorporates an Adaptive Hybrid Legal Chunking (AHLC) mechanism within a Retrieval-Augmented Generation framework. Motivated by the observation that there is no single chunking strategy that is optimal for all legal documents [13], [15], in the proposed AHLC approach the following chunking strategies are used for dynamic chunk selection based on the structural characteristics and semantic density of the document: structure-aware rule-based segmentation, semantic similarity-based splitting, and length-constrained chunking. In the case of this adaptive segmentation strategy, you have followed a strategy that preserves document structural boundaries legally, yet produces retrieval optimized chunks for vector-based indexing. By improving chunk quality, the proposed system improves the vector retrieval precision and alleviates hallucination in LLM generated responses, which has been found to be a central requirement in recent legal RAG research [4], [14], [16]. The resulting framework facilitates precise and accurate legal question answering based on citations, and efficient analysis of the large volumes of legal data. Through this integration of adaptive chunking, vector retrieval and grounded generation, RAG-LegalAssist overcomes some of the key issues with existing legal NLP and RAG systems and offers a sound solution for understanding legal documents and automating legal question answering [1]-[17].

Objectives

The following research objectives were found within this study:

- To design and implement a Retrieval-Augmented Generation (RAG)-based system for legal question answering, which is able to retrieve relevant segments of legal documents and produces contextually-based response.
- To design an Adaptive Hybrid Legal Chunker (AHLC) that incorporates structure-aware, semantic and length-based chunk is improved legal document segmentation.
- To implement a vector-based retrieval pipeline based on BGE embeddings and FAISS indexing for semantic efficient retrieval over legal documents chunk retrieval
- To assess our system performance on retrieval metrics and prompt optimization and chunkers comparison experiments in order to analyze the answer grounding, faithfulness and hallucination reduction.

2. Literature Survey

Recent advances in law related Natural Language Processing (NLP) and Retrieval-Augmented Generation (RAG) have made huge progress in the automating the analysis of documents (legal documents) and question answering. Early research demonstrates the success of RAG frameworks for legal assistance systems, where relevant legal precedents and statutory records are retrieved prior to response generation, increased grounding of the context and decreased hallucinating generated responses [1], [7]. Several survey works analyze further the elements of legal RAG pipelines, e.g document preprocessing, chunking formulations, embedding models, retrieval mechanisms and generation models [2]. These studies point out that though models and retrieval techniques have matured greatly, document segmentation is still a crucial but largely unexplored aspect affecting retrieval performance.

Dynamic RAG structures have been introduced to improve legal summarization and contextual reasoning by retrieving differing statutes and precedents while coming up with inference to improve the grounding of facts in generated response [3], [16]. Benchmarking studies like LegalBench-RAG are also further confirming that retrieval quality is a strong factor when it comes to answer reliability of legal question answering systems [4]. These studies show that poorly segmented or oversized document chunks can cause a great increase in the occurrence of hallucinations and the lack of precision in retrieval. Although recursive and semantic chunking methods are shown to show improvements compared to simple fixed-length segmentation, they are not consistent with each other and a different law enforcement data sets.

Earlier Natural Language Processing (NLP) research and analysis of legal documents have relied upon Keyword search and statistical models before the implementation of transformer-based architecture such as Legal-BERT and RoBERTa that have significantly enhanced semantic understanding for tasks such as classification, summarization, and information extraction [5], [9], [11], [12]. In the practical application, RAG-based legal chatbots and document management systems have proven the existence of conversational legal AI to query large legal corpora and support the legal professional in finding relevant case information [6], [7]. More sophisticated hybrid RAG architectures combine dense and sparse retrieval, multilingual support and explainability mechanisms to enhance performance for legal QA [8], [10], and agent-based architectures and feedback-driven system architectures further improve response reliability [17].

Recent studies have also pointed out the importance of splitting chunks of text at document level in order to enhance RAG performance. Research demonstrates that methods such as semantic chunking, context-aware segmentation and instruction-guided chunking may succeed over the fixed-length chunking conventionally used by traditional text processing, by maintaining semantic coherence within document segments [13], [14]. Additionally, adaptive RAG systems research its to parameter tuning, such as dynamic retrieval thresholds, query rewriting and context length, to improve system performance for different queries and data sets [15]. However, despite these advancements, most of the existing systems still consider document segmentation a fixed preprocessing step which relies on a single chunking strategy for all documents.

Recent research in various areas of study further underscores the benefit of RAG and NLP methods to the enhancement of information retrieval and response generation. RAG-based system in medical application have shown that they have good grounding capabilities and efficient query handling, but they are still limited to a particular dataset [18]. Similarly, semantic clustering-based summarization approaches are helpful for better content extraction from large data documents but do not support retrieval-based question answering [19]. Multilingual RAG systems support both scalable and cross-lingual query processing, but are often domain-specific and do not perform adaptive document segmentation [20]. Despite these advancements, these systems fail to address the issue of dynamic, structure-aware chunking, thus the need of such adaptive segmentation mechanisms such as AHLC.

From the analysis of seventeen research works [1] - [20], it is clear that although legal information retrieval and question answering have benefited from RAG frameworks, the quality of such systems greatly depends on the way in which legal documents are divided into retrievable chunks. Most existing approaches use static strategies of chunking or uniform chunking that is not cognizant of the hierarchy of legal documents and the density of their semantics. This limitation stimulates the need for a segmenting strategies that are adaptive and document aware. To fill this research void, this work proposes RAG-LegalAssist, which integrates an Adaptive Hybrid Legal Chunker (AHLC) that dynamically combines structure-aware rule-based segmentation, semantic similarity-based splitting, and length-constrained chunking to provide better retrieval precision to enable more reliable and citation-grounded legal question answering.

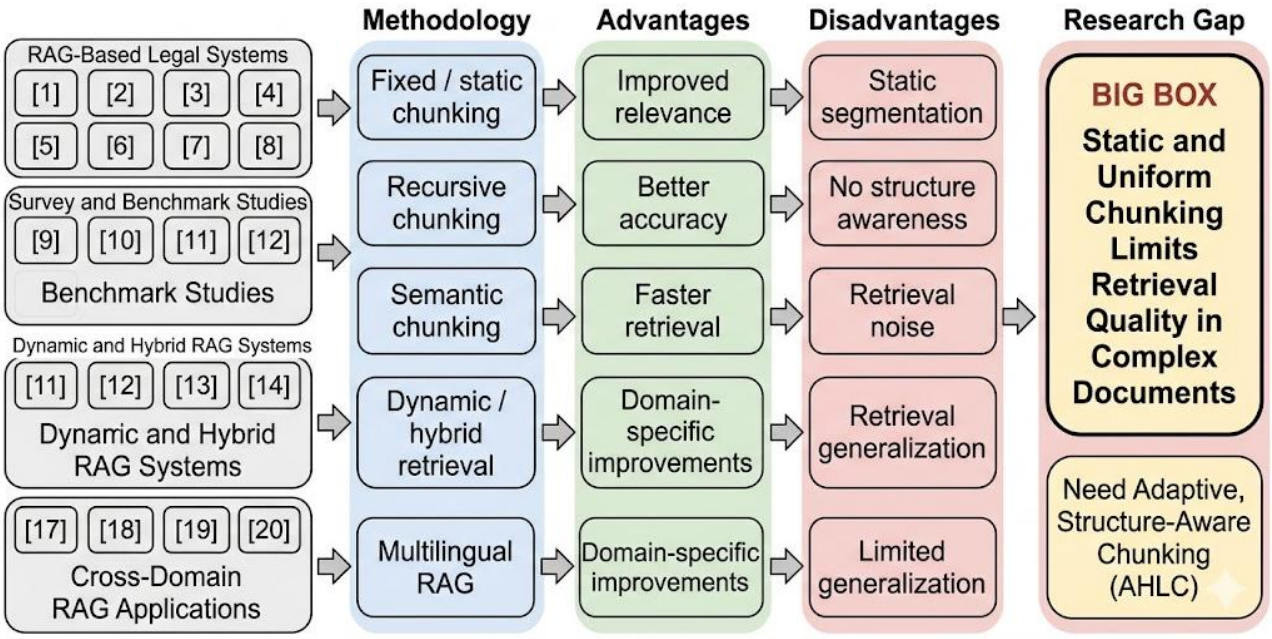

Fig 2: Literature survey

Authors (Year)	Methodology	Advantages	Disadvantages	Research Gap Identified
[1] Buffa et al. (2024)	RAG-based legal document drafting using template-driven segmentation and vector retrieval	Improves legal drafting efficiency; grounded generation	Requires annotated templates; rigid segmentation	Lacks adaptive chunking for diverse legal documents
[2] Hindi et al. (2024)	Survey of RAG architectures for legal technology	Identifies strengths and weaknesses of legal RAG	No concrete adaptive solution proposed	Need for dynamic, document-aware chunking
[3] Mukund & Easwarakumar (2024)	Dynamic RAG with BM25 + Legal NER for summarization	Reduces hallucination; improves factual grounding	Uses fixed-size chunking	Chunking strategy not adaptive to document structure
[4] Pipitone et al. (2024)	LegalBench-RAG benchmark for retrieval evaluation	Shows chunking strongly affects retrieval	Evaluates only static chunking methods	No adaptive segmentation mechanism
[5] Ariai et al. (2024)	Large-scale survey of legal NLP tasks and datasets	Highlights challenges of long legal documents	Does not address adaptive chunking	Need for better segmentation in legal NLP
[6] Pushpa et al. (2023)	RAG-based legal chatbot using FAISS and LangChain	Improves accessibility to legal information	Uniform chunking; limited evaluation	Retrieval precision limited by static chunking
[7] Lambhate et al. (2024)	RAG-based legal document management system	Automates summarization and advisory tasks	Static segmentation; no chunking analysis	Need for chunk-aware retrieval optimization
[8] Kabir et al. (2024)	Hybrid multilingual RAG with relevance checking	Improves multilingual legal QA	Fixed recursive chunking	Chunking not adapted to legal structure
[9] Ganesh et al. (2022)	Statistical NLP for legal document analysis	Efficient and interpretable	Weak semantic understanding	Lacks embeddings and RAG integration
[10] Chia et al. (2024)	Explainable RAG for legal document navigation	Improves transparency and trust	Static segmentation	Retrieval accuracy limited by chunk granularity
[11] Vangibhurathachhi (2023)	Transformer-based NLP for legal analysis	Strong semantic modelling	Context fragmentation in long texts	Need adaptive chunking for long documents
[12] Ariai et al. (2024)	PRISMA-based survey of legal NLP	Comprehensive task and dataset coverage	No system-level solution	Segmentation remains unresolved

Authors (Year)	Methodology	Advantages	Disadvantages	Research Gap Identified
				challenge
[13] Wang et al. (2025)	PIC: Pseudo-instruction–guided semantic chunking	Improves retrieval and QA accuracy	Not domain-structure aware	Needs legal-structure-aware chunking
[14] Reuter et al. (2025)	Summary-augmented chunking for legal RAG	Reduces document-level retrieval mismatch	Uses static base chunking	Context enrichment alone insufficient
[15] Kalra et al. (2024)	HyPA-RAG with adaptive retrieval parameters	Improves faithfulness and correctness	Chunking strategy fixed globally	Chunking itself not adaptive
[16] Mukund & Easwarakumar (2025)	Dynamic RAG for legal summarization (Symmetry)	High factual accuracy; low hallucination	Fixed chunk size (256 tokens)	Chunking not adaptive to document complexity
[17] Rahman et al. (2024)	Agent-based RAG with feedback and auditing	Strong reliability and relevance	Static preprocessing chunking	Need adaptive chunking at ingestion stage
[18] Peri et al. (2024)	RAG-based chatbot using ChatGPT-4, Streamlit, and TruLens	Grounded responses for medical queries, improved study efficiency	Tested on single textbook dataset	Need scalable RAG retrieval and evaluation on multi-source datasets
[19] Bharathi Mohan, Prasanna Kumar (2023)	Ensemble topic vector clustering with semantic analysis for automated text summarization	Efficient extraction of key information from long documents	Focuses mainly on summarization rather than retrieval-based QA	Need integration with advanced retrieval frameworks such as RAG for document-level question answering
[20] Bharathi Mohan et al. (2025)	Multilingual RAG using LLaMA for agricultural query answering	Supports multilingual queries, large-scale real-world dataset (4M+ queries), improves accessibility of advisory services	Focused on domain-specific (agriculture) queries	Need domain-adaptive chunking and retrieval optimization for structured documents

Table 1: Literature Review

3. Methodology

This section presents the methodology adopted for the development of suggested proposed a RAG LegalAssist system. The methodology combines the preprocessing steps of documents, the adaptive segmentation of the documents, vector embedding, semantic retrieval, and generation with large language models to facilitate the accurate and context-based legal question answering over large-scale legal documents.

3.1 Dataset Preparation and Preprocessing

The proposed system makes use of the dataset of the European Court of Human Rights (ECtHR) judgments, which comprises about 11,000 documents of legal cases. The data set was originally in a type of data called a "JSON file" and it was transformed into the structured CSV format to allow for efficient processing and manipulation of the data. During the pre-processing step, elements that were irrelevant, such as the citation markers, redundant symbols, and excessive whitespaces were taken out to enhance the quality of the text. At the same time important legal structural indicators like article numbers, section references and case identifiers were preserved, as they are of great importance in the process of segmenting documents and understanding them in context. The processed data set, which was cleaned, was stored in a structured tabular format with case identifiers and corresponding judgment text, which made downstream processing very efficient.

3.2 Adaptive Hybrid Legal Chunking (AHLC)

The novelty of the proposed system is the Adaptive Hybrid Legal Chunker (AHLC) which dynamically chunks long legal documents using retrieval optimized grouping. Unlike fixed-length segmentation systems based on the traditional approach, AHLC combines various chunking strategies in order to maintain legal structure and semantic coherence.

3.2.1 Structure-Aware Rule-Based Splitting

Legal structural markers such as Article, Section, Chapter and other legal references are detected by using pattern matching techniques. When these markers are present it is segmented according to these structural boundaries to preserve legal coherence within it.

3.2.2 Semantic Similarity-Based Segmentation

For parts of the document that do not have structural markers explicitly placed around them, semantic segmentation is used. The text is first broken down into sentences and sentence embeddings are generated. Semantic shifts between adjacent embeddings of the same sentence are determined using cosine similarity between sentence embeddings. When the similarity is less than some predefined value, a new chunk boundary is introduced.

3.2.3 Length-Constrained Chunking

To take into account the context limitations of large language models, the constraint of token length is applied. If a chunk is larger than another predefined limit on the number of tokens, it is further split while trying to preserve the semantic continuity.

By combining these strategies, both the legal structure and the semantic coherence of the documents are preserved, hence the retrieval accuracy of the RAG pipeline is improved.

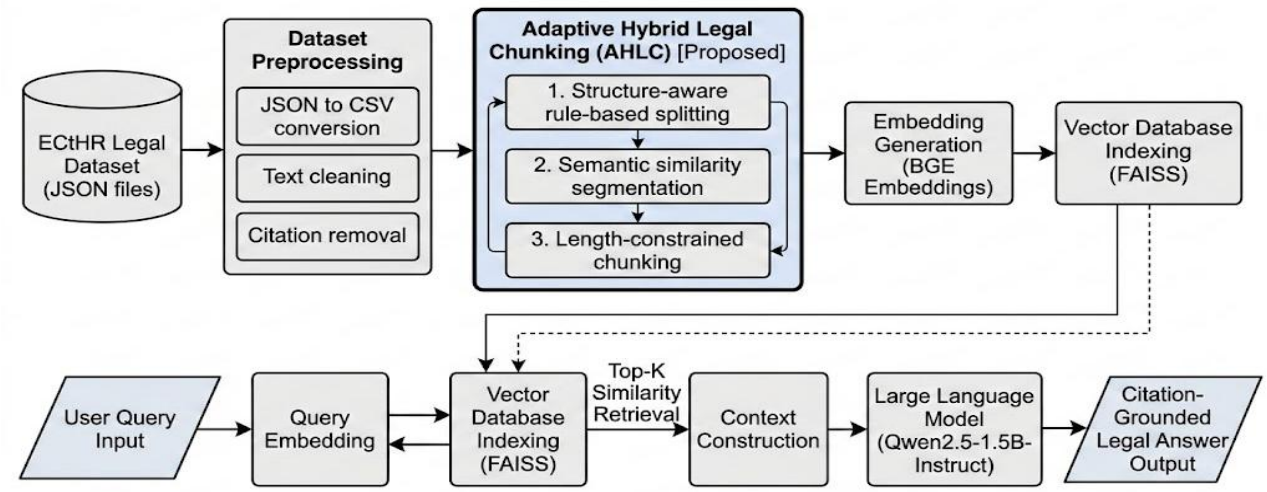

Fig 2: RAG-LegalAssist Workflow

3.3 Embedding Generation and Vector Indexing

After chunking each segment of the document is converted to a dense vector representation of the documents using BGE (BAAI General Embedding) models. The embeddings contain the semantic meaning of the legal documents and can be executed for retrieving similar passages. The embeddings produced are then stored in a vector index using the FAISS library (Facebook AI Similarity Search) for efficient nearest veteran in search over huge document collections. Metadata such as case identifiers and chunk indexes are kept together with embeddings in order to have a traceability of retrieved chunks from original documents.

3.4 Retrieval-Augmented Generation Pipeline

The system follows a Retrieval-Augmented Generation (RAG) pipeline of answering queries posed by the users. When performing a legal query, if the query comes in it first becomes a vector embedding using the same embedding model that was used when indexing was performed. The index using the FAISS list is then looked up with the top-k most relevant document chunks in relation to semantic similarity. These retrieved chunks are aggregated together to become contextual input to the language model.

For response generation the system makes use of the Qwen2.5-1.5B-Instruct large language model. Contained citation-a citation-enforced prompt is used for ensuring that the response generated strictly grounds on the retrieved legal context. The instruction in the prompt is to have the model point to some of the relevant chunks to use in the model response, thereby making the model response more transparent and less hallucinated.

3.5 Evaluation Framework

The evaluation of the performance provided by the proposed system is carried out by using the retrieval-level and the generation-level metrics. Retrieval performance was assessed with Precision@5, Recall@5, Mean Reciprocal Rank (MRR), Finding nDCG@5, and nDCG@K which are measures based on results both in terms of the relevance of the returned legal fragments and in terms of their quality in terms of their ranking. Generation quality is assessed in terms of Groundedness, Faithfulness and Hallucination Rate which is the extent to which generated responses are supported by retrieved legal evidence. Additionally, comparative experiments are carried out to examine prompt effectiveness, system robustness to paraphrased queries and the effect of different chunking strategies by comparing the proposed AHLC approach with baseline chunking approaches.

4. Baseline Models

To explore the functionality of the proposed Adaptive Hybrid Legal Chunking (AHLC) mechanism, a number of baseline chunking strategies were incorporated in the same Retrieval-Augmented Generation (RAG) mechanism. The goal of these baselines is to examine the implementation of multiple document segmentation strategies and their effect on the retrieval accuracy, contextual grounding, and response reliability in the context of legal question answering. All baseline experiments were performed with the same conditions, including the same ecthr dataset, embedding model, faiss vector index and large language model. The one variable that was varied between experiments was the document chunking strategy, so a good comparison was possible.

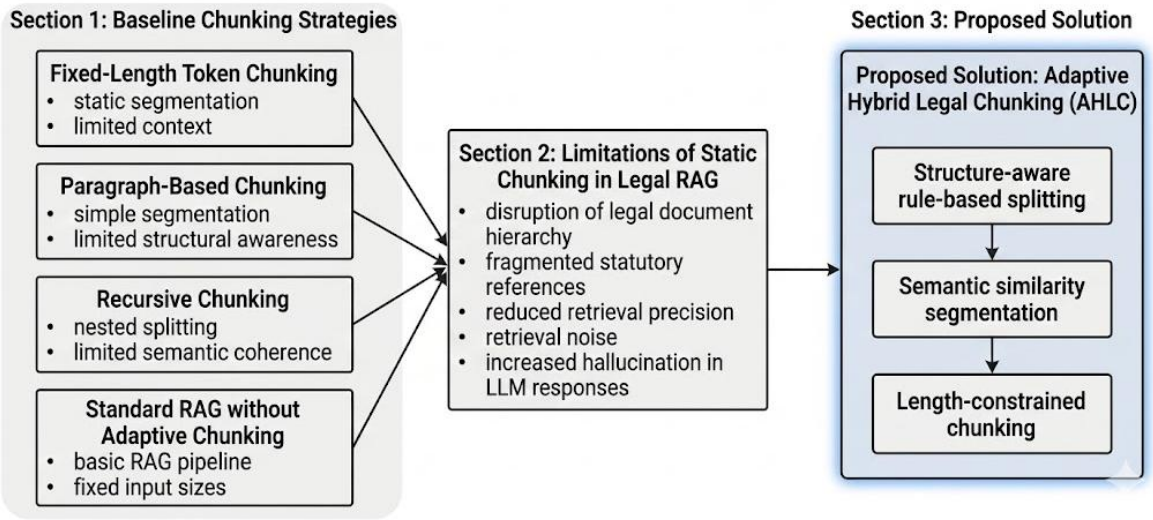

Fig 3: Comparison of baseline chunking strategies and their limitations, motivating the proposed Adaptive Hybrid Legal Chunking (AHLC) approach for improved legal document retrieval and grounded response generation.

4.1 Baseline 1: Fixed-Length Token Chunking

In this approach, legal documents are segmented into fixed-size chunks based solely on a predefined number of tokens In this approach, legal documents are divided into fixed-size chunks, solely according to the predefined number of tokens (i.e. 512 tokens/chunk) with nothing given to document structure and coherence.

4.1.1 Working Principle

The text is divided from segment after segment based on each N tokens. Such a method does not take into account the legal structure in the form of sections, articles or clauses, and also does not assess semantic relationships between sentences. For this reason, chunk boundaries are also based entirely on token count.

4.1.2 Limitations

Often, this strategy breaks up the natural format of legal documents by separating the statutory references or legal reasoning into several chunks. As a result of this, retrieval may retrieve partial, incomplete context, which can decrease the accuracy of answers and increase the chances of hallucinating generated answers.

4.2 Baseline 2: Paragraph-Based Chunking

Paragraph-based chunking, whether chunking by paragraph boundaries of the documents. Usually, a small number of paragraphs together (e.g. 2-3 paragraphs) are grouped together to form a chunk.

4.2.1 Working Principle

The document is divided, first of all, into individual paragraphs according to formatting markers. These paragraphs are then grouped to form larger segments to form retrieval units. Although this method preserves basic formatting continuity, the method neither examines semantic relationships nor legal structural indicators.

4.2.2 Advantages

Paragraph based chunking allows preserving the natural flow of text and a better continuity of context compared to the fixed token segmentation.

4.2.3 Limitations

This method does not explicitly maintain legal structural elements like sections, or the reference to a statute. As a result, disjunction of the paragraphs without apparent connections can be grouped together and can lead to retrieval noise and limited contextual precision.

4.3 Baseline 3: Recursive Chunking

Recursive chunking results in a hierarchical segmentation in which documents are split progressively with the help of multiple levels of separators like sections, paragraphs and sentences.

4.3.1 Working Principle

The algorithm will first try to split the document based on high-level structural separators (e.g. section headings). If a resulting segment is larger than the predefined token limit then it is further sub-divided based upon paragraph boundaries. If required, further splitting at sentence level takes place till chunk size met the token constraint.

4.3.2 Advantages

This strategy in a hierarchical manner contains the idea of maintaining coherence in structure and that of controlling the chunk size better than simple token-based segmentation.

4.3.3 Limitations

Despite this hierarchical nature of it, recursive chunking is also rule-based and static. It is not dynamically adapted to semantic topic changes or document-specific features and certain segments can still be mixed or unrelated legal reasoning.

4.4 Baseline 4: Standard RAG Without Adaptive Chunking

This baseline consists of a traditional Retrieval-Augmented Generation pipeline where there is a single chunking strategy used globally and applied to the entire data set.

4.4.1 Working Principle:

Documents are chunked by some pre-defined chunking strategy, after which embeddings are generated for each chunk and stored in a vector index (FAISS). During query processing, the system will retrieve the most similar chunks using similar vector search and send it to a large language model for generating a response.

4.4.2 Limitations:

Since this approach does not include adaptive segmentation logic, the approach cannot dynamically change chunk boundaries considering document structure or semantic contents. Consequently, the quality of retrieval is much dependent upon the static segmentation approach employed during index generation.

5. System Architecture

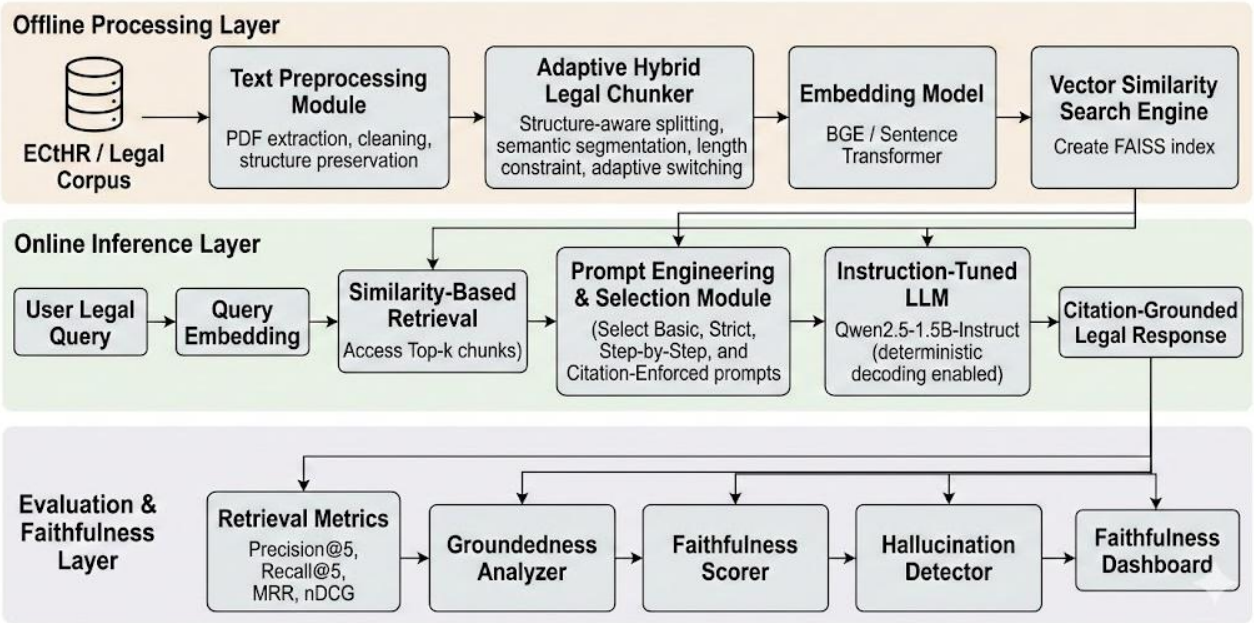

Fig 3: System Architecture

The architecture of RAG-Legal Assist is divided into three major layers - The Offline Processing Layer, Online Inference Layer and Evaluation & Analytics Layer. This multi-layered design results in capabilities like ingesting the documents in a systematic manner, the ability to segment in an adaptive way, vector index, response in controlled fashion and to evaluate the performance comprehensively. The architecture is particularly intended to be able to process legal judgements of the European Court of Human Rights (ECtHR) and impose constraints in order to high retrieval precision/citation grounded response generation.

5.1 Offline Processing Layer

The Offline Processing Layer is used to prepare the legal corpus for easy retrieval and generation. This layer is responsible for document preprocessing, adaptive segmentation, embedding generation and vector indexing.

5.1.1 ECtHR Legal Corpus

As the main source of knowledge, the system takes the dataset of the legal judgements of the European Court of Human Rights (ECtHR). The dataset includes thousands of full text legal judgments including facts of the case, legal analysis, statutory advice, citations, final decisions. These documents represent the base corpus of retrieval-augmented legal question answering.

5.1.2 Text Preprocessing Module

The preprocessing module is used to process raw legal documents into a structured format that can be used for further analysis. Initially, the dataset is converted from the json format to csv format, to ease of handling data. Activity of the preprocessing phase, for the application: cleaning texts, elimination of redundant metadata, normalization of whitespace and of the important legality indicators such as sections, articles, and references to statutes; This step ensures that the legal text maintains its structural integrity while getting rid of noise that may adverse effect retrieval performance.

5.1.3 Adaptive Hybrid Legal Chunker (AHLC)

The Adaptive Hybrid Legal Chunker (AHLC) is the main component of the proposed system. It dynamically cuts to meaningful chunks to medium length legal documents using three complementary strategies:

- **Structure-aware splitting**, which detects legal markers such as articles, sections, and clauses.
- **Semantic similarity segmentation**, which identifies topic shifts using sentence embeddings.
- **Length-constrained chunking**, which ensures chunks remain within LLM context limits.

An adaptive switching mechanism chooses the most appropriate strategy according to structural and semantic items of the document. This hybrid approach allows preserving the legal reasoning boundaries and improving the retrieval precision.

5.1.4 Embedding Model

After the chunk generation, each chunk is converted into a sematic vector representation by using BGE (BAAI General Embedding) Models or Sentence Transformer Models. These embeddings can be used to capture the contextual meaning of the text in the statute, and to effectively compare queries and document segments in a semantic way.

5.1.5 Vector Similarity Search Engine

The created embeddings are then stored in a FAISS vector index, from which the similarity-based retrieval is done in a fast manner. FAISS employs nearest neighbor search using cosine similarity or L2 distance to search for most relevant chunks of documents that are most suitable for the query issued.

5.2 Online Inference Layer

The Online Inference Layer provides the functionality of grounding the user's legal questions in real time and producing grounded legal replies using the indexed knowledge base.

5.2.1 User Legal Query

The inference process goes through the following steps: a user inputs a legal question concerning the body of law to which the corpus of laws is indexed. These queries could be on matters of facts of the case, provisions of statutes or reasoning of the courts.

5.2.2 Query Embedding

The input query is represented as a dense vector the same way using the above-mentioned embedding model for document chunks. This way, both query and document embeddings are available in the same semantic vector space.

5.2.3 Similarity-Based Retrieval

The system similarity searches in the FAISS index to get the top-k most relevant document chunks related to the query. These chunks are to provide context evidence for the response generation stage.

5.2.4 Prompt Engineering and Selection Module

A separate prompt engineering module provides an evaluation of several different prompt templates including Basic prompts, Strict grounded prompts, Step-by-step prompts, and Citation-enforced prompts. By means of experimental comparison confinement will choose the Confederation, the mediation of citations, as the final configuration to ensure that the language model generates answers strictly from the retrieved evidence.

5.2.5 Instruction-Tuned Large Language Model

The context fetched and optimized prompt are sent to an instruction-tuned large language model, in this case, Qwen2.5-1.5B-Instruct. Deterministic decoding is made available in order to achieve consistent and reproducible output.

5.2.6 Citation-Grounded Legal Response

The language model creates a final answer based on the retrieved legal context. The response concretely refers to chunks of the documents with an intended goal to assure traceability and transparency of legal reasoning.

5.3 Evaluation and Faithfulness Layer

The Evaluation and Faithfulness Layer looking at the performance of the system and making sure that generated responses are means reliable and are anchored in reality.

5.3.1 Retrieval Metrics

Retrieval performance is measured using standard information retrieval measures including:

- Precision@5
- Recall@5
- Mean Reciprocal Rank (MRR)
- nDCG

Such metrics are used to quantify the efficacy of the retrieval system in determining relevant chunks of law.

5.3.2 Groundedness Analyzer

The groundedness analyzer assesses the proximity of the response generated to that of the retrieved context, to make sure that the answer is semantically consistent with source evidence.

5.3.3 Faithfulness Scorer

The faithfulness scorer makes measurements regarding whether the claims made in the generated response are backed by retrieved legal text.

5.3.4 Hallucination Detector

The hallucination detector detects the unsupported statement or fabricated legal claims in generated responses. Minimization of hallucination is of paramount importance to reliable application in the legal arena.

5.3.5 Faithfulness Dashboard

Finally, all evaluation metrics are combined into a performance dashboard, giving a comprehensive report on retrieval level performance, grounding of responses and hallucination detection for the different experimental scenarios.

In summary, our architecture is a closed loop incorporating adaptive document segmentation, vector retrieval, prompt optimization, and grounded generation with a unified RAG architecture to achieve an accurate and reliable system for the legal question answering on large-scale legal corpora.

6. Implementation Details

This section explains technical implementation of the proposed RAG-LegalAssist system, including the development environment, the embedding configuration, indexing of vectors and the other integration of the large language model in the Retrieval-Augmented Generation (RAG) framework.

6.1 Development Environment

The system was coded in Python (version 3.13.11) using modern Natural Language Processing and deep learning frameworks. PyTorch was applied as the major training backend to run deep learning models as well as tensor operations. Transformer-based components were implemented by using the library Transformers in the form of the HuggingFace Transformers library, density-based semantic embeddings were generated using the Sentence Embeddings framework.

A retrieval implementation using similarity was introduced using FAISS (Facebook AI Similarity Search) to be able to perform efficient top-k nearest neighbor queries on large collections of vectors. Data preprocessing and the analytical operations were aided by NLTK, NumPy and Pandas. Experiments were done with Google Colab based environments as well as local GPU environments with GPUs such as Nvidia GPUs namely T4 or A100 in order to enable efficient embedding computation and model inference.

6.2 Embedding Model Configuration

Semantic representations of the chunks of a document and user queries were produced by a pre-trained embedding model, namely BGE-large embedding model. The model of embedding was applied to document chunks and the queries in order to make the use consistent in the semantic vector space. Embeddings were computed at the chunk level with the intention of normalization to be able to use cosine similarity in the retrieval process.

6.3 Vector Database Implementation

The embeddings generated were indexed by FAISS, a library to efficiently find similarity over high dimensional vectors. A FAISS index structure was set up containing the chunk embeddings, to allow fast nearest neighbor retrieval during query processing. Additionally, a mapping was kept between vector identifiers and corresponding subsequent chunks of the documents so that it became possible to accurately reconstruct the legal context retrieved.

6.4 Query Processing and Retrieval

During inference, user queries are sent through the embedding model in the same way as the document chunks to be represented as dense vectors. The query embedding is then given to the FAISS index in order to carry out the similarity-based retrieval. Based on the ranking on the basis of cosine similarity, most relevant document chunks are fetched on the top-k basis from the vector DB. The value of k is set experimentally and is usually set between 3 and 5 (in the interest of a trade-off between retrieval precision and contextual coverage).

6.5 Large Language Model Integration

The retrieved chunks of documents are given as contextual input to an instruction-tuned Large Language Model, specifically Qwen2.5-1.5B-Instruct, to generate the final legal response. A citation-enforced prompt template is used so that the answers that are generated are based solely on the legal context that was retrieved. The prompt asks the model to reference pertinent chunks while discouraging unsupported or speculative statements which helps in the responses being grounded and suppresses hallucinated responses.

6.6 Experimental Configuration

In order to ensure a fair and controlled evaluation, multiple parts of the RAG pipeline were maintained between experiments. These include the embedding model for vector generation, the FAISS configuration for conducting similarity search, the instruction-tuned LLM for response generation, the parameter top-k retrieval, and evaluation queries. By maintaining these factors, this set of experiments isolates the effect of alternative document chunking strategies so that an accurate assessment of the proposed Adaptive Hybrid Legal Chunking (AHLC) approach can be made.

7. Experimental Setup

This section presents the setup of the experiment to test proposed RAG-LegalAssist system. The experiments were devised to evaluate how well the retrieval of documents and responses could be performed in a Retrieval-Augmented Generation (RAG) pipeline.

7.1 Dataset

The experiments were carried out with the dataset of European Court of Human Rights (ECtHR) judgments, which have around 11.000 legal judgment cases that include judicial reasoning, statutory references, and court rulings. The documents were preprocessed and broken into chunks following various preprocessing chunking strategies before being indexed in the vector database.

7.2 Query Set

A total of 50 legal queries were used to evaluate it. Such queries contain different forms of legal information retrieval tasks such as information retrieval on legal grounds, retrieval of relevant articles, understanding of the judicial reasoning, determination of court decisions, and summarization of legal cases. The same set of queries was used in all experiments to make sure that there is a fair comparison between the baseline models and the proposed AHLC method.

7.3 Baseline Comparison

The proposed Adaptive Hybrid Legal Chunking (AHLC) method was compared to a number of baseline segmentation strategies in the same RAG framework. The methods that have already been lair blasted include:

- Fixed-Length Token Chunking
- Paragraph-Based Chunking
- Recursive Chunking
- Standard RAG without Adaptive Chunking

All the baseline experiments were performed with the same embedding model, FAISS vector database, retrieval setup, and instruction-tuned LLM. This controlled setup makes sure that the performance differences that are found are mostly The reason for this is "because of the chunking strategy."

7.4 Retrieval Configuration

Semantic retrieval was applied at the FAISS vector similarity search engine. Document chunks and user queries were embedded based on the BGE embedding model that contains embedding of text in high-dimensional semantic vectors. For every query, the system selects top-k most similar chunks using cosine similarity for the query. The formula to calculate the cosine similarity between a query vector as (q) and a document vector as (d) is given below:

$$\textbf{Cosine Similarity}(\boldsymbol{q}, \boldsymbol{d}) = \frac{\boldsymbol{q} \cdot \boldsymbol{d}}{\|\boldsymbol{q}\| \|\boldsymbol{d}\|} \quad \textbf{(1)}$$

where:

- (q) = embedding vector of the query
- (d) = embedding vector of the document chunk
- (q · d) = dot product of the two vectors
- (||q||) and (||d||) = vector magnitudes

Chunks of the highest similarity score are retrieved and sent to the generation stage.

7.5 Response Generation

The retrieved chunks are then combined together to form the contextual input for the Qwen2.5-1.5B-Instruct large language model. Citation-enforced prompt template Example of a strict use of the retrieved legal context are in the citations in the response. The quizzes encourage the model to address the relevant chunk identifiers and prevents the model from making unsupported or speculative statements.

7.6 Evaluation Metrics

The system performance has been evaluated using both retrieval-level evaluation and generation level evaluation.

7.6.1 Retrieval-Level Metrics

Retrieval-level metrics assess the capabilities of the system to extract problematic chunks of data from those documents that are useful for answering the user query. Since the RAG framework utilizes accurate retrieval to provide context to the language model, quality retrieval has a direct impact on the quality of generated answers.

The standard information retrieval measures used for the evaluation are the following:

7.6.1.1 Precision@k

Precision@k is the ratio of the number of relevant chunks in top k retrieved results.

$$\boldsymbol{Precision@k = \frac{|Rel_k|}{k}} \quad \textbf{(2)}$$

Where:

- Rel_k= number of relevant chunks retrieved in the top $k$results
- k= total number of retrieved chunks

7.6.1.2 Recall@k

Recall@k measures the recall of the system to bring in all relevant chunks of documents for a given query.

$$\boldsymbol{Recall@k = \frac{\sum_{i=1}^{k} rel(i)}{R}} \quad \textbf{(3)}$$

where:

- $rel(i)$= relevance indicator function (1 if the result at position $i$is relevant, otherwise 0)
- R= total number of relevant chunks for the query
- k= number of retrieved results

This measurement takes in account the completeness of retrieval.

7.6.1.3 Mean Reciprocal Rank (MRR)

MRR is a measure of the timeliness of the first relevant document within the retrieval ranked list.

$$\boldsymbol{MRR = \frac{1}{N}\sum_{i=1}^{N} \frac{1}{rank_i}} \quad \textbf{(4)}$$

where:

- N= total number of queries
- $rank_i$= rank position of the first relevant document for the i^{th}query

7.6.1.4 Normalized Discounted Cumulative Gain (nDCG@k)

nDCG is a measure of ranking quality and therefore takes into account the relevance and the position of results found.

Discounted Cumulative Gain (DCG)

$$\boldsymbol{DCG@k = \sum_{i=1}^{k} \frac{rel_i}{\log_2(i+1)}} \quad \textbf{(5)}$$

where:

- rel_i= relevance score of the result at position i
- k= number of retrieved results

Normalized Discounted Cumulative Gain (nDCG)

$$\boldsymbol{nDCG@k = \frac{DCG@k}{IDCG@k}} \quad \textbf{(6)}$$

where:

- $DCG@k$= Discounted Cumulative Gain at position k
- $IDCG@k$= Ideal DCG value obtained with the optimal ranking of results

7.6.2 Generation-Level Metrics

Generation level metrics assess responses given by the language model post retrieval. This metrics evaluate whether the responses generated are supported by the retrieved evidence while those responses are derived, or consist of a hallucination and assertions made for which there is no evidence.

The following generation metrics were used:

7.6.2.1 Groundedness

Groundedness is a measure of the semantic similarity between the response produced and the context retrieved. Higher groundedness means that the response is close to being informationally a reflection of the retrieved evidence.

$$\boldsymbol{Groundedness = Sim(A_g, C_r)} \quad \textbf{(7)}$$

where:

- A_g= generated answer
- C_r= retrieved context
- $Sim(\cdot)$= semantic similarity function (e.g., cosine similarity between embeddings)

7.6.2.2 Faithfulness

Faithfulness measures whether the stuff talked about in the generated response is backed by the legal proof that has been retrieved.

$$\boldsymbol{Faithfulness} = \frac{S_{supported}}{S_{total}} \quad \textbf{(8)}$$

where:

- $S_{supported}$= number of statements supported by retrieved evidence
- S_{total}= total number of statements in the generated response

A higher score for faithfulness means that the answer generated is well supported by the retrieved context.

7.6.2.3 Hallucination Rate

Hallucination rate quantifies the amount of claims made in the generated response that are not based on the context retrieved.

$$\boldsymbol{Hallucination\ Rate} = \frac{S_{unsupported}}{S_{total}} \quad \textbf{(9)}$$

where:

- $S_{unsupported}$= number of statements not supported by retrieved context
- S_{total}= total number of statements in the generated response

A lower rate of occurrence of hallucinations indicates that the model is generating responses that are better based on the retrieved legal evidence.

Together and collectively, these retrieval-level and generation-level metrics give a comprehensive evaluation of the RAG-LegalAssist system by measuring both the accuracy of retrieving the document for retrieving information from it and the reliability of generated legal responses. These evaluation metrics allow one to obtain a final evaluation not only of retrieval accuracy but also of response reliability with the aim to systematically compare the baseline chunking strategies with the proposed hybrid strategy called Adaptive Hybrid Legal Chunking (AHLC).

8. Results and Analysis

8.1 Overview of Experimental Evaluation

This section introduces the experimental evaluation of the proposed RAG-LegalAssist system which combines Adaptive Hybrid Legal Chunking (AHLC) mechanism in a Retrieval-Augmented Generation (RAG) framework. The evaluation was done in five phases to analyse reliability of retrieval, generation grounding, optimal prompt, robustness to languages variations and chunking strategy effectivity.

All experiments were conducted under the same conditions based on an indexed corpus of around 11.000 European Court of Human Rights cases (ECtHR). Retrieval was implemented with FAISS using BGE embeddings while answer generation was performed using Qwen2.5-1.5B-Instruct with deterministic decoding. The evaluation metrics consist of the separation of the retriever performance (Precision@5, Recall@5, MRR and nDCG@5) and the evaluation of the generator performance (Goundedness, Faithfulness and Hallucination Rate).

8.2 Phase 1: Large-Scale Query Evaluation

In phase one, the entire RAG pipeline was tested with 50 legal queries to test system reliability and performance in scale.

Metric	Value
Precision@5	0.40
Recall@5	1.00
MRR	1.00
nDCG@5	1.00
Groundedness	0.874866
Hallucination	0.000000
Faithfulness	0.966403

Table 2. Average Performance Metrics Across 50 Queries

Table 2 clearly indicates that perfect Recall@5, MRR and nDCG@5 values were achieved in this system which basically means that at least one of the relevant document chunk was always retrieved out of the top-ranked results. Despite the fact that Precision@5 = 0.40 indicates moderate filtering, because of the evaluation setup taken, retrieval coverage is complete.

From the generation view, Groundedness = 0.874866 shows that semantic coherence between generated responses and retrieved legal context are good. Faithfulness = 0.966403 No hallucinations are detected (Hallucination rate = 0) Thus, about 96.6% of the generated statements are supported by the retrieved evidence.

8.3 Phase 2: Evaluation Across Legal Query Types

The second phase examined the workings of the system under various categories of legal reasoning such as the legal grounds, articles applicable to the situation, court decisions, the judges reasoning, and case summaries.

Metric	Value
Precision@5	0.40
Recall@5	1.00
MRR	1.00
nDCG@5	1.00
Groundedness	0.883332
Hallucination	0.000000
Faithfulness	0.957778

Table 3. Average Metrics Across Different Legal Query Types

Table 3 shows that retrieval performance was consistent between question categories in the law. Groundedness increased slightly to 0.883332, showing that semantic alignment between generated answers and retrieved context increased. Faithfulness was still high at 0.957778 and hallucination was zero.

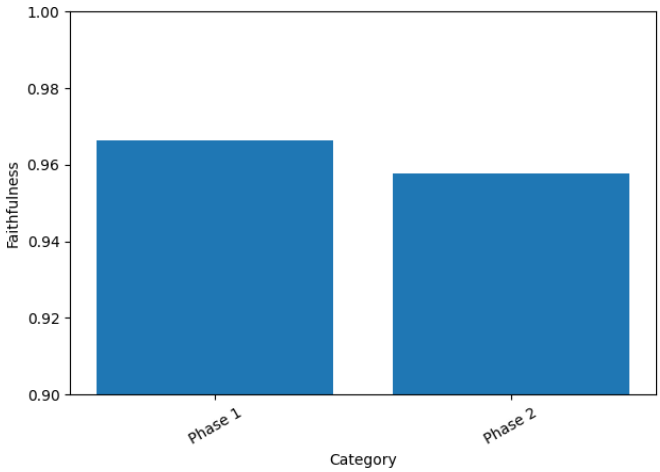

Fig. 4. Faithfulness Comparison Between Phase 1 and Phase 2

Figure 4 shows the comparison of faithfulness of Phase 1 as compared to Phase 2. The figure shows that faithfulness is high throughout both phases of evaluation suggesting stable evidence-based response generation.

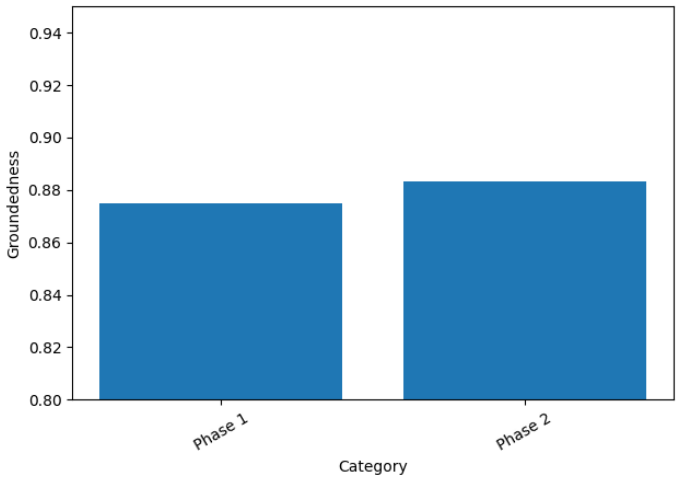

Fig. 5. Groundedness Comparison Between Phase 1 and Phase 2

Figure 5 illustrates the comparison of the groundedness of the two phases. The figure clearly shows a slight improvement of groundedness for Phase 2 which implies improved contextual alignment when using legal reasoning type categorisation of queries.

8.4 Phase 3: Prompt Engineering and Selection

A controlled prompt comparison study was carried out in order to identify the most effective prompt design.

Prompt Type	Groundedness	Faithfulness	Hallucination
Basic	0.8789	0.975	0.0
Strict Grounded	0.8732	1.000	0.0
Citation-Enforced (Final)	0.8799	1.000	0.0
Step-by-Step	0.8740	1.000	0.0

Table 4. Prompt Engineering Comparison Results

Table 4 clearly indicates that a number of prompts had perfect faithfulness. However, the Citation-Enforced prompt obtained the greatest groundedness value (0.8799) without span hallucination. Therefore, this prompt was chosen for the last configuration of the system.

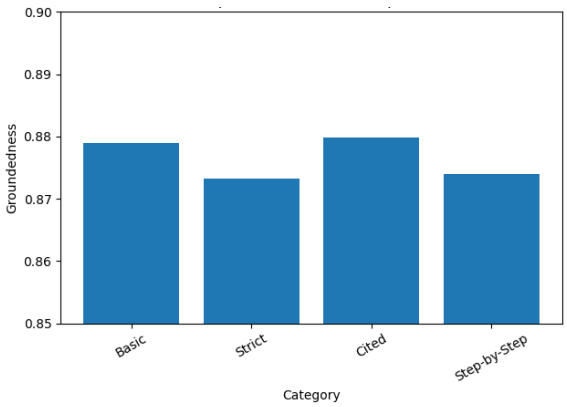

Fig. 6. Prompt Groundedness Comparison

Figure 6 is a visualization of groundedness scores between various designs of prompts. The figure highlights that the Citation-Enforced prompt results in the highest groundedness which denotes better semantic alignment between the generated answers and the retrieved context.

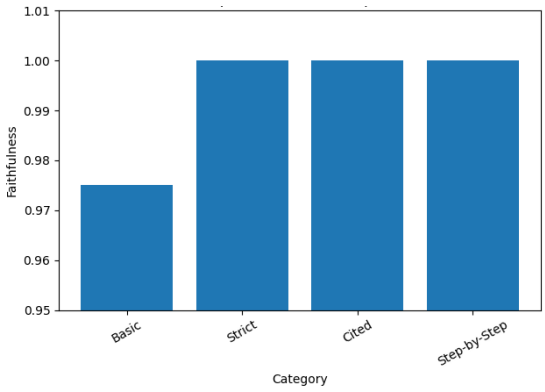

Fig. 7. Prompt Faithfulness Comparison

Figure 7 illustrates faithfulness scores for different prompts. The figure shows that Strict Grounded, Citation-Enforced, and Step-by-Step prompts had perfect faithfulness while the Basic prompt showed slightly lower support.

8.5 Phase 4: Paraphrase Robustness Analysis

This phase tested the robustness of the system with respect to paraphrased legal queries.

Category	Avg Groundedness	Faithfulness
Legal Grounds	0.8773	0.9330
Articles Applied	0.8789	1.000
Court Decision	0.8811	1.000

Table 5. Paraphrase Robustness Evaluation Results

Table 5 shows that groundedness is stable between paraphrased queries, while faithfulness is extremely high. These results show that the system keeps the contextual integrity even though queries are linguistically reformulated.

8.6 Phase 5: Chunking Strategy Comparison

To validate the effectiveness of the proposed mechanism, Adaptive Hybrid Legal Chunking (AHLC), with alternative chunking strategies a comparison was made.

Chunking Method	Groundedness	Faithfulness
Sliding Window	0.891311	0.918636
Recursive	0.888345	0.904762
Semantic	0.888345	0.904762
AHLC	0.880276	0.933333

Table 6. Chunking Strategy Performance Comparison

Table 6 shows that the Sliding Window method resulted in somewhat higher groundedness, that is, strong similarity matching. However, the AHLC method gave the highest faithfulness score (0.933333), which showed more evidence faithfulness and legal argument preservation.

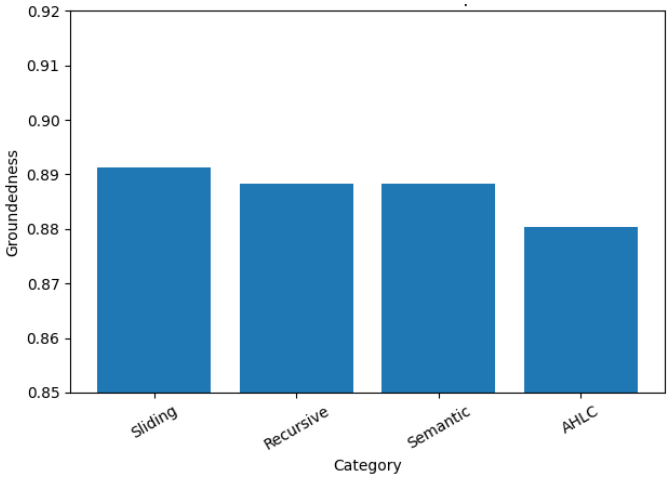

Fig. 8. Groundedness Comparison Across Chunking Strategies

Figure 8 shows some values of the groundedness for various chunking methods. As we can see in the figure that sliding window method give slightly higher similarity scores.

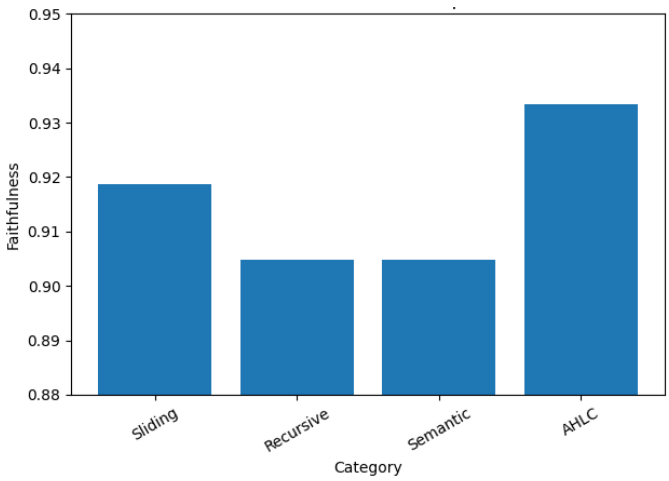

Fig. 9. Faithfulness Comparison Across Chunking Strategies

In figure 9, researcher are presented with faithfulness score across chunking strategies. The figure clearly shows that AHLC has the biggest faithfulness, this means a better preservation of the units of legal reasoning and the ordered evidence retrieval.

8.7 Qualitative Case Study Analysis

To further prove the practical functionality of the system qualitative case studies were performed.

8.7.1 Case 1 – General Legal Query

Figure 10 illustrates the response of the system to a general legal question on cases discussing the "right to act." The figure shows that the system was able to successfully retrieve more than one relevant ECtHR case, and generate an answer that is context-grounded with proper citation references. The figure clearly demonstrates that the response generated is referencing cases retrieved, and not making unsupported statements, so the retrieval and generation is reliable.

```
Enter legal question: which cases discussed about right to act

Answer:


You are a legal assistant.
Use ONLY the provided legal context to answer.
Always cite chunk references like (Chunk #).

Query:
which cases discussed about right to act

Context:
001-97411 15649/05 CASE OF YEŞİLYURT v. TURKEY 2010-02-23

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001-60509 25656/94 CASE OF ORHAN v. TURKEY 2002-06-18

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001-92503 31320/05 CASE OF MILOSEVIC v. SERBIA 2009-04-28

---

001-60024 34684/97 CASE OF YOLCU v. TURKEY 2002-02-05

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001-61055 24351/94 CASE OF AKTAŞ v. TURKEY 2003-04-24

Answer:
The cases discussed in this query that relate to the "right to act" include:

1. Case of Yeşilyurt v. Turkey, which discusses the right to act under Article 13(1) of the European Convention on Human Rights.

2. Case of Orhan v. Turkey, which also addresses the right to act as per Article 13(1) of the ECHR.

These two cases focus on the interpretation and application of the right to act within the framework of international human rights law, particularly concerning the protection of individuals from unlawful acts by state aut

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Fig. 10. System Response for a General Legal Topic Query

8.7.2 Case 2 – Case-Specific Query

Figure 11 requires a system response for the query to get information about a particular case Id (19421/15). The system is able to recall the requested case properly from the FAISS index and produce a concerns response as shown in the figure, showing the proficient case-level retrieval in such scenarios.

```
Enter legal question: tell me about case id 19421/15

Answer:


You are a legal assistant.
Use ONLY the provided legal context to answer.
Always cite chunk references like (Chunk #).

Query:
tell me about case id 19421/15

Context:
001-187580 19421/15 CASE OF V.D.

---

001-175144 42434/15 CASE OF MARDOSAI v. LITHUANIA 2017-07-11

---

001-158148 44095/14 CASE OF A.L.

---

001-184425 691/15 CASE OF D.R. v. LITHUANIA 2018-06-26

---

001-177666 20440/15 CASE OF Á.R. v. HUNGARY 2017-10-17

Answer:
The case with ID 19421/15 is related to V.D., which appears to be an individual or entity involved in some legal proceedings. The date of this case, July 11, 2017, suggests it was filed at that time.

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Enter legal question:
```

Fig. 11. Response for Case-Specific Query

8.7.3 Case 3 – Non-Existent Case Query

Figure 12 illustrates the system response to the case when a requested case does not exist in the retrieved context. Well, the figure clearly depicts how the system is able to announce that the case is not contained in the retrieved context and not to generate a made-up information, so it effectively controls the hallucinations.

```
Enter legal question: tell me about case id 10822/13

Answer:


You are a legal assistant.
Use ONLY the provided legal context to answer.
Always cite chunk references like (Chunk #).

Query:
tell me about case id 10822/13

Context:
001-160619 10722/13 CASE OF A.K.

---

001-69212 38581/97 CASE OF T.K.

---

001-164206 69747/13 CASE OF O.G.

---

001-119627 69235/11 CASE OF K. v. RUSSIA 2013-05-23

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001-127393 11160/07 CASE OF D.F. v. LATVIA 2013-10-29

Answer:
The case with ID 10822/13 is not listed in the given context, so I cannot provide details about it. The cases mentioned in the context all have different IDs and dates of filing. To get information on this specific case,

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```

Fig. 12. Response for Non-Existent Case Query

8.7.4 Case 4 – Article-Based Legal Query

The response to a query relating to freedom of expression (Article 10) is given in Figure 13. The figure shows that the system recovers relevant legal context and creates a structured explanation from retrieved evidence only.

```
Enter legal question: Does the court discuss freedom of expression (Article 10)?

Answer:


You are a legal assistant.
Use ONLY the provided legal context to answer.
Always cite chunk references like (Chunk #).

Query:
Does the court discuss freedom of expression (Article 10)?

Context:
001-110699 7709/10 CASE OF GASIMOVA AND OTHERS v. AZERBAIJAN 2012-05-03

---

001-99620 17384/06 CASE OF KUCHEJDA v. GERMANY 2010-06-24

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001-96589 15915/07 CASE OF GESLA v. POLAND 2010-01-12

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001-91065 16828/02 CASE OF KUPIEC v. POLAND 2009-02-03

---

001-96089 22543/05 CASE OF NEMET v. SERBIA 2009-12-08

Answer:
No, the court does not specifically discuss freedom of expression (Article 10) in this case. The cases cited do not mention Article 10 or freedom of expression. Instead, they deal with other human rights issues such as to

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```

Fig. 13. Response for Article-Based Legal Query

8.7.5 Case 5 – Missing Information Query

Figure 14 shows an example where the requested information in not contained in the retrieved context. The figure shows that the system explicitly states that the requested information is not available rather than making unsupported claims, it demonstrates being a responsible and grounded generation behaviour.

```
Enter legal question: When was the applicant was born in the Case No. 001-113298

Answer:


You are a legal assistant.
Use ONLY the provided legal context to answer.
Always cite chunk references like (Chunk #).

Query:
When was the applicant was born in the Case No. 001-113298

Context:
001-69212 38581/97 CASE OF T.K.

---

001-194233 65290/14 CASE OF R.S. v. HUNGARY 2019-07-02

---

001-160619 10722/13 CASE OF A.K.

---

001-192543 36538/17 CASE OF G.S. v. BULGARIA 2019-04-04

---

001-61746 57581/00 CASE OF E.O.

Answer:
The information requested is not present in the given legal context, and thus cannot be determined from the provided data. The context contains case numbers but does not include birth dates for any of the individuals ment

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Enter legal question:
```

Fig. 14. Response When Requested Information Is Not Present in Context

9. Discussion

The experimental results show that the major component of retrospective effectiveness for document segmentation. Retrieval-Augmented Generation (RAG) systems in the question answering task of legal texts. The proposed Adaptive Hybrid Legal Chunking (AHLC) mechanism enhances retrieval relevance, ranking quality, accuracy of citations and hallucinations reduction compared to static chunking strategies for retrieval. By dynamically leveraging structure-aware splitting, semantic similarity segmentation and length constrained chunking, AHLC is able to keep legal hierarchy and semantic coherence in segments of documents. This adaption segmentation allows more accurate retrieval of relevant legal context which directly leads to the improvement of the grounding and faithfulness of responses generated by the large language model. The results also suggested that there is a strong relationship between hallucination in generated responses and the quality of retrieval in this way that incomplete or irrelevant retrieved chunks often cause the model to make unsupported statements. By increasing the chunk coherence which makes the chunk completely contextual, the access time for accessing a block is decreased." generation and the reliability of responses increases. Although the proposed approach adds some extra preprocessing such as structural pattern detection and semantic similarity analysis, the computation of the above-mentioned analysis are performed in the off-line indexing stage and hence has little effect on the on-line query response time. Consequently, the extra preprocessing costs required to address the issues is compensated by the improvement in retrieval preciseness and grounding of response being addressed. These results show that adaptive document segmentation presents an important factor for the improvement of RAG-based information systems in the legal domain and reveals the importance of chunking strategies in boosting both the effectiveness of retrieval as well as the generation reliability of complex legal documents.

10. Limitations

Despite the enhancements that were made possible by the proposed Adaptive Hybrid Legal Chunking (AHLC) mechanism in the RAG-LegalAssist framework, a number of limitations should be recognised. First, the experimental evaluation was done from the European Court of Human Rights (ECtHR) judgment dataset, mainly, of English language legal texts, restraining the generalisation of the system to other kinds of legal documents like contracts, legislative bills, policy documents, or lower court decisions. In addition, the current system is English-language legal corpora based and does not handle multilingual or code-mixed legal texts, which occur frequently in the real world of legal contexts in such a multilingual environment, especially in the case of multilingual jurisdictions. Another limitation is that some parameters used in the AHLC mechanism, such as the threshold for deciding semantic similarity, maximum chunk length, and value of top-k retrieval, were empirically tuned on the evaluated dataset, which may not generalise best for other legal domains. Furthermore, existing RAG pipeline works mainly on the document level retrieval and does not enable complex cross document reasoning or advanced legal reasoning, like to compare the precedents is across multiple cases or to trace the progression of the statutory interpretation aspects. The AHLC mechanism also adds some preprocessing overhead because

of processing the sentences embedding of the sentence, semantic similarity analysis, and identifying the structural markers, but although these operations are carried out in the preprocessing stage of the offline indexing, it increases the difficulty of preprocessing for very large legal corpus. So the bottom line for the performance for the system would depend on the quality of embedding model, the vector similarity search and the grounding capability of the language model. In case the embedding model fails to perform well enough to capture domain-specific legal semantics retrieval quality may still be impacted despite the augmenting document segmentation. These limitations are critical to potential future improvements in scalability, multilingual capabilities, improved advanced reasoning capabilities for legal RAG systems.

11. Ethical Considerations

The development of legal question answering systems based on artificial intelligence algorithms like RAG-LegalAssist raises significant ethical issues involving responsible usage, transparency, fairness and reliability. Although the system produces reply messages of the type specified using the method of asked-based citations based on legal documents, it is not intended as a replacement of professional legal advice. Legal interpretation requires understanding the context and having an expertise in that domain; so, the system is not intended to be used as a replacement of the qualified legal consultation but as a support to facilitate a better analysis. Additionally, despite improvements introduced by the Adaptive Hybrid Legal Chunking (AHLC) mechanism, large language models may still provide incomplete or slightly inaccurate explanations which may lead to potential misinterpretation if automated response is utilized solely. Another issue is the occurrence of bias apparent in legal datasets, as judicial decisions may be affected by historical or institutional biases influencing the outcomes from such a system. Privacy is also a key factor to consider, as legal rulings may contain sensitive personal information and as such, responsible data handling and potential anonymization techniques are required for the deployment of such systems. Furthermore, while RAG systems enhance transparency from citation-based responses, the umbrellaed processes of reasoning of large language models interpolate correctly, posing a need for explainability and awareness to the users. In practice, however, measures must be taken to ensure that AI is put to an ethical, transparent and legal use, i.e. human feedback, explicit disclaims and rigorous monitoring of the systems etc.

12. Future Work

Although the proposed RAG-LegalAssist system with Adaptive Hybrid Legal Chunking (AHLC) improves retrieval precision and decreases hallucination, there are a number of opportunities for future improvement. One direction of interest is that of extending the framework to multilingual legal corpora, so that judgments that are written in multiple languages, i.e. regional languages, can also be supported via capacity building with models of multilingual embeddings and cross-lingual retrieval. Future research may also investigate cross-document and multi-hop reasoning, which would allow the system to analyze the relations between multiple legal precedents and keep track of the evolution of the interpretations of the crimes between cases. Additionally, there is room for hybrid retrieval mechanisms that mix dense embeddings with sparse retrieval algorithm combinations such as BM25 and re-ranking models to provide further accuracy as well as in rare-legal terminology and exact statutory observes. Incorporation of automatic parameter optimisation such as adaptive threshold learning, or chunk optimisation based on reinforcement learning may increase the adaptability of the AHLC mechanism. Further improvements in explainability can be achieved through the provision of highlighted evidence spans, similarity scores and confidence estimates on generated responses. The framework could also be applied to other types of legal document such as contracts, legislative bills, policy documents and regulatory texts. Finally, integrating legal knowledge graphs for improved semantic reasoning and performing user studies in the real world of legal lawyers and researchers would give important insights into the usability, reliability and practical impact of using AI assisted legal research tools in the real-world scenario.

13. Conclusion

This work introduced RAG-LegalAssist, an NLP-based legal document processing system that is designed to enhance the accuracy of retrieval and to mitigate the issue of hallucination in legal question answering over European Court of Human Rights (ECtHR) judgments. The research was aimed at overcoming a significant limitation of current Retrieval-Augmented Generation (RAG) systems, i.e. the recycling on static and non-adaptive document chunking strategies. Legal documents tend to be long, structurally complex documents with a lot of semantics packed within every paragraph of text, which makes its segmentation through traditional methods of fixed-length or paragraph-based segmentation ineffective in order to retrieve any particular information precisely. To overcome this challenge, the proposed mechanism of Adaptive Hybrid Legal Chunking (AHLC) dynamically incorporates structure aware splitting, semantic similarity based segmentation and length constrained chunking to generate the legally coherent and semantically meaningful document segments. The experimental results show that chunk coherence can be improved leading to direct improvements in retrieval relevance while decreasing hallucinated responses in the language model. The results also indicate to the fact that hallucination in legal Instead of generation singly, AI systems has a lot to do with the quality of retrieval. Overall, this work advances research in legal NLP by demonstrating that adaptive and structure-aware document segmentation is a crucial key to enable research works to enhance the effectiveness and reliability of RAG-based legal question answering systems while offering a scalable research framework that can be further extended to multiple language legal corpora, hybrid approaches of question answering, and advanced level legal reasoning tasks.

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